

Ivan Evdokimov, Ph.D.

Manchester, UK | ivevdm@gmail.com | linkedin.com/in/ivan-evdokimov | github.com/ivan020

Professional Summary

Software Engineer (4+ years) building secure, maintainable APIs and web systems with Python, and delivering full-stack products with TypeScript/React front ends and PostgreSQL-backed back ends. Experienced designing and operating CI/CD pipelines, cloud infrastructure (AWS), and containerised, event-driven services, with 95%+ model accuracy delivered on sensitive production datasets. Builds and applies AI-assisted development workflows in production, writing clear prompts, critically reviewing AI-generated code, and taking ownership of its correctness and security. Comfortable gathering requirements directly from academic and public-sector stakeholders and translating them into production specifications; has taught and mentored 100+ students and colleagues.

Experience

Software Engineer – Full Stack & MLOps, University of Essex — Colchester, UK Nov 2025 — Present

- Built GitHub Actions/Vercel CI/CD pipelines that automate testing for an AI-assisted development workflow, enabling weekly releases across a multi-team platform.
- Deployed end-to-end MLOps infrastructure with continuous production monitoring for a multi-team platform, reducing deployment time and improving release reliability.
- Developed a Java mobile app integrating the OpenAI API for 20 field technicians, automating inspection reports and saving 3 weeks/year of manual data collection.

Software Engineer – Backend & ML, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed domain-specific PyTorch and scikit-learn models using Hugging Face, vLLM, and Ollama for automated metadata classification and disclosure risk assessment on a public data portal, achieving 95%+ accuracy on sensitive social science datasets.
- Deployed and orchestrated Dockerised event-driven ML retraining pipelines using AWS Lambda and Step Functions with infrastructure-as-code practices, adding drift/accuracy monitoring that auto-triggers retraining and cuts manual intervention by 50%.
- Built ETL pipelines to extract, transform, and query PostgreSQL-hosted social science datasets of 100K+ records as model inputs for automated metadata classification, improving throughput and classification accuracy at scale.
- Optimized statistical disclosure control algorithms by migrating from R to C++ using bitmask techniques, achieving 11x performance improvement and enabling real-time processing of datasets with 500k+ records.
- Gathered and refined requirements directly with academics, project managers, and public-sector clients, translating non-technical needs into production ML system specifications.

Research Officer & ML Instructor, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime 8x for collaborations with other universities in the UK and the USA.
- Delivered lectures and labs in Data Structures, C/C++, and Machine Learning to 100+ undergrad and postgrad students, achieving an average student feedback score of 4.8/5.

Projects

Chat-LLM github.com/ivan020/ChatLLM

- Built a self-hosted RAG pipeline over a Raspberry Pi-hosted Ollama LLM using LangChain, pgvector, and all-MiniLM-L6-v2 embeddings, indexing past stream transcripts to allow new viewers query past content in real time via Twitch chat.
- Optimised the TinyLlama-1.1B-Chat-v1.0 model for low-latency inference under constrained hardware.

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Built a Dockerised real-time portfolio tracker in Go with buy/sell mechanics, a PostgreSQL (Supabase) backend recording all transactions, and a live leaderboard — using Twitch chat as the user interface.
- Developed a TypeScript React frontend with real-time portfolio visualisation, interactive performance graphs, and transaction history, serving 120+ monthly users.

MTG Card Reader ivan020.github.io/MtgRecognizer/

- Built a Dockerised Golang REST API that uses OCR to extract text from Magic: The Gathering card photos, identifies the card via LLM, and returns real-time price data from a PostgreSQL-backed catalogue.
- Developed a TypeScript React frontend with card filtering, deck management, and real-time price display, deployed on Vercel.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Developed a PyTorch transfer learning framework for Bayesian model averaging in financial forecasting, first-authored a peer-reviewed CS/ML journal publication, and co-authored 3 international conference papers; released an open-source package with external adoption.

University of Essex, MSc in Financial Econometrics Oct 2020 — Sep 2021

- Sole-authored a modular C++ agent-based simulation modelling household consumption, employment, and borrowing behaviour under dynamic and negative interest rates, with an extensible architecture supporting additional firm and household modules.